Overview

Ned2 is a six-axis collaborative robot, based on open-source technologies. It is intended for education, research and Industry 4.0.

The Ned2



The Ned2 made with alumnium structure will meet your requirements in terms of **robustness** and **repeatability (0.5 mm)**.

Ned2 is based on **Raspberry PI 4**, a **high performance 64-bit ARM V8 processor** and **4gb of RAM**.

Discover a robot with **increased performances**, thanks to **new servo motors featuring Silent Stepper Technology**, significantly reduces the noise level of the robot.

Ned2 is a **collaborative robot based on Ubuntu and ROS** (Robot Operating System) Noetic, an open-source solution created for robotics. Through ROS, **Ned2 has multiple libraries**

allowing you to conceive many programs, from the most simple to the most complex ones, responding then, in a flexible way, to your needs.

Ned2 tools technical specifications



Mounting a tool on Ned2 is made easy with our **EasyConnect system**. Simply plug your tool, connect its cable to the control panel, and it is ready to use.

Ned2's package includes a Custom Gripper. Its standard jaws can manipulate small objects, and you can 3D print your own custom jaws.

Ned2 can also be used with:

- The Large Gripper which has the ability to grab larger objects while keeping the ability to close entirely;
- The Adaptive Gripper which allow Ned2 to grab non-standard objects with complex shapes (eg. an egg);

- The Vacuum Pump to grab objects with plane and non-porous surfaces;
- The Electromagnet that is useful to manipulate metallic objects, from one to many (eg. screws, bolts...).

Ned2's Ecosystem is designed to let you reproduce advanced use cases of industry 4.0 :

- The Vision Set gives Ned2 the ability to detect objects and select the ones you want to interact with. It comes with a camera and objects of different shapes and colors to allow you to study topics such as image processing, machine learning and Artificial Intelligence thanks to the contribution of vision.
- The Conveyor Belt is the key add-on to prototype industry 4.0 production lines.

LED ring indications

Color	Condition	Operating mode	Description	Troubleshooting
White **Breath**	Normal condition		Robot is booting	N/A
White to blue **Breath**	Normal condition		Starting the robot	N/A
Slow blue **Chase**	Normal condition	Manual mode - Calibration	Calibration required	Press the Custom button, or launch a calibration
Blue **Snake**	Normal condition	Manual mode - Calibration	Calibration in progress	N/A
3 yellow **Flashing**	Normal condition	Manual mode - Calibration	Calibration warning	N/A
Solid orange	Warning	Manual mode	One or more robot axes are outside the motion limits	Return the axes to within the limits and start a calibration.

Color	Condition	Operating mode	Description	Troubleshooting
1 white **Breath** lasting 2 seconds then 6 blinks in white	Normal condition	Manual mode	Shutdown in progress	N/A
Flashing red	Emergency	Manual mode	Motor error	Check the motor error codes in the “Hardware Status” panel of Niryo Studio. Contact the after-sales service if it persists, as there may be defective fan.
Solid white	Normal condition		Update in progress	N/A
Breath blue	Normal condition	Manual mode - FreeMotion	FreeMotion activated	N/A
Breath green	Normal condition	Manual mode	FreeMotion disabled / Ready to start program	N/A
Solid green	Normal condition	Manual mode - Programming	Program running	N/A
Chase green	Normal condition	Manual mode - Programming	Program paused	A short press on the button to resume, a long press on the top button to cancel

Color	Condition	Operating mode	Description	Troubleshooting
				the current program
Chase orange	Warning	Manual mode - Programming	Error in the program	Check the logs on Niryo Studio and Launch a new action to clear this state
Solid red	Emergency		Fatal error	Check the logs on Niryo Studio and restart the robot
Flashing orange	Warning		A collision is detected	Perform a new action to remove the warning and check if the robot has collided with an obstacle
Flashing purple	Normal condition	Manual mode	Login to Niryo Studio	N/A
1 **Flashing** white	Normal condition	Manual mode – programming	Saving a position	N/A

Pattern description

LEDs are turned off



Solid : Set the whole LED Ring to the same color at once



Flashing : Flashes a color according to a frequency



Alternate: The different colors are alternated one after the other.



Chase : Movie theater light style chase animation



Rainbow : Draws rainbow that fades across all LEDs at once.



Breath : Variation of light intensity to imitate breathing.



Snake : Luminous snake that turns around the LED Ring.



> In order to program the LED ring of the robot , please refer to APIs for Sounds and lights

---

Made with Docsie.io